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TASK : Multi-Tier VPC Setup with Public & Private Subnets

Create a secure VPC with public and private subnets, where:

- Public subnet hosts a bastion host (jump server) and an Application Load Balancer (ALB).
- Private subnet hosts EC2 application servers.
- Only the bastion host can SSH into private instances.
- Outbound internet access for private instances is through a NAT Gateway.

Expected Outcome:

Public subnet: ALB + Bastion.

Private subnet: EC2 app servers.

Secure access: Only bastion can SSH into private EC2.

Private EC2 instances can reach the internet via NAT.

Users can access the app only through ALB.

Step 1: Create VPC

- Created VPC **MultiTierVPC** with CIDR 10.0.0.0/16.
- Tenancy: **Default**.

The screenshot displays the AWS VPC console. In the 'Your VPCs' section, a table lists the VPCs. The 'MultiTierVPC' is highlighted, showing its ID, state, and CIDR. Below this, the 'Details' tab for 'vpc-040104d4cae02a0fa / MultiTierVPC' is shown. It contains various configuration details organized into sections: VPC ID, State (Available), Block Public Access (Off), DNS hostnames (Disabled), DNS resolution (Enabled), Tenancy (default), DHCP option set, Main network ACL, IPV6 CIDR (Network border group), Default VPC (No), Network Address Usage metrics (Disabled), IPV4 CIDR, Route 53 Resolver DNS Firewall rule groups, Main route table, IPV6 pool, and Owner ID.

Step 2: Create Subnets

- **PublicSubnet-1a**: 10.0.1.0/24 in AZ us-east-1a.
- **PrivateSubnet-1a**: 10.0.2.0/24 in AZ us-east-1a.

us-east-1.console.aws.amazon.com/vpconsole/home?region=us-east-1#subnets:sort=desc:tag:Name

Subnets (2/8) Info

Find subnets by attribute or tag

	Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR	IPv6 C
☒	PublicSubnet-1a	subnet-051194440aa334ca3	Available	vpc-040104d4cae02a0fa Multi...	Off	10.0.1.0/24	-
☒	PrivateSubnet-1a	subnet-043aa613841f0f133	Available	vpc-040104d4cae02a0fa Multi...	Off	10.0.2.0/24	-
☐	-	subnet-021f7341603804494	Available	vpc-0610fd9fedfd36a33	Off	172.31.16.0/20	-
☐	-	subnet-064c2e2616cd0f429	Available	vpc-0610fd9fedfd36a33	Off	172.31.32.0/20	-

Subnets: subnet-051194440aa334ca3, subnet-043aa613841f0f133

Step 3: Create & Attach Internet Gateway

- Created Internet Gateway **MultiTierIGW**.

igws | VPC Console

us-east-1.console.aws.amazon.com/vpconsole/home?region=us-east-1#igws:

Internet gateways (1/2) Info

Find internet gateways by attribute or tag

	Name	Internet gateway ID	State	VPC ID	Owner
☐	-	igw-0087300ee6e5db57f	Attached	vpc-0610fd9fedfd36a33	253490795695
☒	multiTierIGW	igw-0ce7b64f600251ba5	Detached	-	253490795695

igw-0ce7b64f600251ba5 / multiTierIGW

Details Tags

Details

Internet gateway ID igw-0ce7b64f600251ba5	State Detached	VPC ID -	Owner 253490795695
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- Attached it to **MultiTierVPC**.

The screenshot shows the AWS Management Console for the 'us-east-1' region. The left sidebar displays the 'VPC' dashboard with a list of services including Virtual private cloud, Security, and PrivateLink. The main content area shows the 'Internet gateways (1/2)' page. A table lists two internet gateways: one in an 'Attached' state (igw-0087300ee6e5db67f) and one in a 'Detached' state (igw-0ce7b64f600251ba5). The 'multiTierIGW' gateway is selected. An 'Actions' dropdown menu is open, showing options like 'View details', 'Attach to VPC', 'Detach from VPC', 'Manage tags', and 'Delete internet gateway'. Below the table, the details for the selected gateway 'igw-0ce7b64f600251ba5 / multiTierIGW' are shown, including its ID, state (Detached), VPC ID, and owner.

The screenshot shows the 'Attach to VPC (igw-0ce7b64f600251ba5)' wizard in the AWS Management Console. The 'VPC' section explains that attaching an internet gateway to a VPC enables the VPC to communicate with the internet. Under 'Available VPCs', the VPC 'vpc-040104d4cae02a0fa' is listed and selected. At the bottom, there is a section for the 'AWS Command Line Interface command' and two buttons: 'Cancel' and 'Attach internet gateway'.

Step 4: Create NAT Gateway

- Deployed **Public NAT Gateway** in PublicSubnet-1a.
- Allocated an Elastic IP for internet access.

Attention required on case 175 | VPC | us-east-1

us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#CreateNatGateway:

Search [Alt+S]

United States (N. Virginia) Account ID: 2534-9079-5695 Uddhav Gund

VPC > NAT gateways > Create NAT gateway

NAT gateway settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.
multiTierNATpublic
The name can be up to 256 characters long.

Subnet
Select a subnet in which to create the NAT gateway.
subnet-051194440aa334ca3 (PublicSubnet-1a)

Connectivity type
Select a connectivity type for the NAT gateway.
☒ Public
☐ Private

Elastic IP allocation ID [Info](#)
Assign an Elastic IP address to the NAT gateway.
eipalloc-06b62bd0a484e486c [Allocate Elastic IP](#)

► **Additional settings** [Info](#)

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key
Q Name X

Value - optional
Q multiTierNATpublic X [Remove](#)

[Add new tag](#)

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Attention required on case 175 | NatGateways | VPC Console

us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#NatGateways:

Search [Alt+S]

United States (N. Virginia) Account ID: 2534-9079-5695 Uddhav Gund

VPC > NAT gateways

VPC dashboard < EC2 Global View Filter by VPC

Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet gateways
Carrier gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers [New](#)

Security
Network ACLs
Security groups

PrivateLink and

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NAT gateways (1/1) [Info](#)

Find NAT gateways by attribute or tag

Name	NAT gateway ID	Connectivity...	State	State message	Primary public I...	Primary private I...	Primary network...
multiTierNATpublic	nat-0760d456672d1aebb	Public	Available	-	44.210.184.45	10.0.1.108	eni-0284d01bea7c

nat-0760d456672d1aebb / multiTierNATpublic

[Details](#) [Secondary IPv4 addresses](#) [Monitoring](#) [Tags](#)

Details

NAT gateway ID nat-0760d456672d1aebb	Connectivity type Public	State Available	State message -
NAT gateway ARN arn:aws:ec2:us-east-1:253490795695:natgateway/nat-0760d456672d1aebb	Primary public IPv4 address 44.210.184.45	Primary private IPv4 address 10.0.1.108	Primary network interface ID eni-0284d01bea7c4673 Link
VPC vpc-040104d4cae02a0fa / MultiTierVPC	Subnet subnet-051194440aa334ca3 / PublicSubnet-1a	Created Monday, September 8, 2025 at 11:09:27 GMT +5:30	Deleted -

Step 5: Configure Route Tables

- Public Route Table (PublicRT)
 - Routes:
 - 10.0.0.0/16 → local
 - 0.0.0.0/0 → Internet Gateway (igw-xxxx)
 - Associated with PublicSubnet-1a.

us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#EditRoutes:RouteTableId=rtb-061ecea0aa138ea06

Search [Alt+S]

United States (N. Virginia) Account ID: 2534-9079-5695 Uddhav Gund

VPC > Route tables > rtb-061ecea0aa138ea06 > Edit routes

Edit routes

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	CreateRouteTable
0.0.0.0/0	Internet Gateway	-	No	CreateRoute
	igw-0ce7b64f600251ba5			

Add route

Cancel Preview Save changes

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RouteTables | VPC Console

us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#RouteTables:

Search [Alt+S]

United States (N. Virginia) Account ID: 2534-9079-5695 Uddhav Gund

VPC > Route tables

Route tables (1/3) Info

Find route tables by attribute or tag

	Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC	Owner ID
☐	-	rtb-0aca6e2bac9fc7f01	-	-	Yes	vpc-0610fd9fedfd36a33	253490795695
☐	-	rtb-05c25166fc7f26c87	-	-	Yes	vpc-040104d4cae02a0fa Multi...	253490795695
☒	PublicRT	rtb-061ecea0aa138ea06	-	-	No	vpc-040104d4cae02a0fa Multi...	253490795695

rtb-061ecea0aa138ea06 / PublicRT

Details Routes Subnet associations Edge associations Route propagation Tags

Details

Route table ID	Main	Explicit subnet associations	Edge associations
rtb-061ecea0aa138ea06	No	-	-
VPC	Owner ID		
vpc-040104d4cae02a0fa MultiTierVPC	253490795695		

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- **Private Route Table (PrivateRT)**
 - Routes:
 - 10.0.0.0/16 → local
 - 0.0.0.0/0 → NAT Gateway (nat-xxxx)
 - Associated with PrivateSubnet-1a.

VPC | us-east-1

us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#CreateRouteTable:

Search [Alt+S]

United States (N. Virginia) Account ID: 2534-9079-5695 Uddhav Gund

VPC > Route tables > Create route table

Create route table Info

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

Route table settings

Name - optional

Create a tag with a key of 'Name' and a value that you specify.

PrivateRT

VPC

The VPC to use for this route table.

vpc-040104d4cae02a0fa (MultiTierVPC)

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Q Name

Value - optional

Q PrivateRT

Remove

Add new tag

You can add 49 more tags.

Cancel

Create route table

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VPC | us-east-1

us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#EditRoutes:RouteTableId=rtb-0a0ae79b5022638ae

Search [Alt+S]

United States (N. Virginia) Account ID: 2534-9079-5695 Uddhav Gund

VPC > Route tables > rtb-0a0ae79b5022638ae > Edit routes

Edit routes

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	CreateRouteTable
Q 0.0.0.0/0	NAT Gateway	-	No	CreateRoute
	Q nat-0760d456672d1aebb			

Add route

Cancel

Preview

Save changes

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RouteTables | VPC Console

us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#RouteTables:

Search [Alt+S]

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VPC > Route tables

VPC dashboard

EC2 Global View

Filter by VPC

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

Carrier gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Security

Network ACLs

Security groups

PrivateLink and

CloudShell Feedback

Route tables (2/4) Info

Find route tables by attribute or tag

Last updated 7 minutes ago

Actions Create route table

	Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC	Owner ID
☐	-	rtb-0aca6e2bac9fc7f01	-	-	Yes	vpc-0610fd9fedfd36a33	253490795...
☐	-	rtb-05c25166fc7f26c87	-	-	Yes	vpc-040104d4cae02a0fa Multi...	253490795...
☒	PublicRT	rtb-061ecea0aa138ea06	-	-	No	vpc-040104d4cae02a0fa Multi...	253490795...
☒	PrivateRT	rtb-0a0ae79b5022638ae	-	-	No	vpc-040104d4cae02a0fa Multi...	253490795...

Route tables: rtb-061ecea0aa138ea06, rtb-0a0ae79b5022638ae

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Step 6: Launch Bastion Host (Jump Server)

- Instance Name: **BastionHost**
- AMI: Amazon Linux 2023
- Type: t2.micro
- Subnet: PublicSubnet-1a
- Auto-assign Public IP: **Enabled**
- Security Group **BastionSG**:
 - Inbound: **SSH (22)** from My IP only
 - Outbound: All traffic

Instances | EC2 | us-east-1

(1) Chat | AWS Unofficial | Micro

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#Instances:v=3;\$case=tags:true%5C,client:false;\$regex=tags:false%5C,client:false

Search [Alt+S]

United States (N. Virginia) Account ID: 2534-9079-5695 Uddhav Gund

EC2 > Instances

EC2

Dashboard

EC2 Global View

Events

Instances

Images

AMIs

AMI Catalog

Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

Network & Security

Security Groups

Elastic IPs

Placement Groups

Key Pairs

Network Interfaces

Load Balancing

Load Balancers

CloudShell Feedback

Instances (1/1) Info

Find Instance by attribute or tag (case-sensitive)

All states

Last updated less than a minute ago

Connect Instance state Actions Launch instances

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS
☒	BastionHost	i-0340624a678785006	Running	t2.micro	2/2 checks passed	View alarms	us-east-1a	-

i-0340624a678785006 (BastionHost)

Details Status and alarms Monitoring Security Networking Storage Tags

Instance summary Info

Instance ID i-0340624a678785006

IPV6 address -

Hostname type IP name: ip-10-0-1-173.ec2.internal

Answer private resource DNS name -

Auto-assigned IP address

Public IPv4 address 100.24.28.80 | open address

Instance state Running

Private IP DNS name (IPv4 only) ip-10-0-1-173.ec2.internal

Instance type t2.micro

VPC ID

Private IPv4 addresses 10.0.1.173

Public DNS -

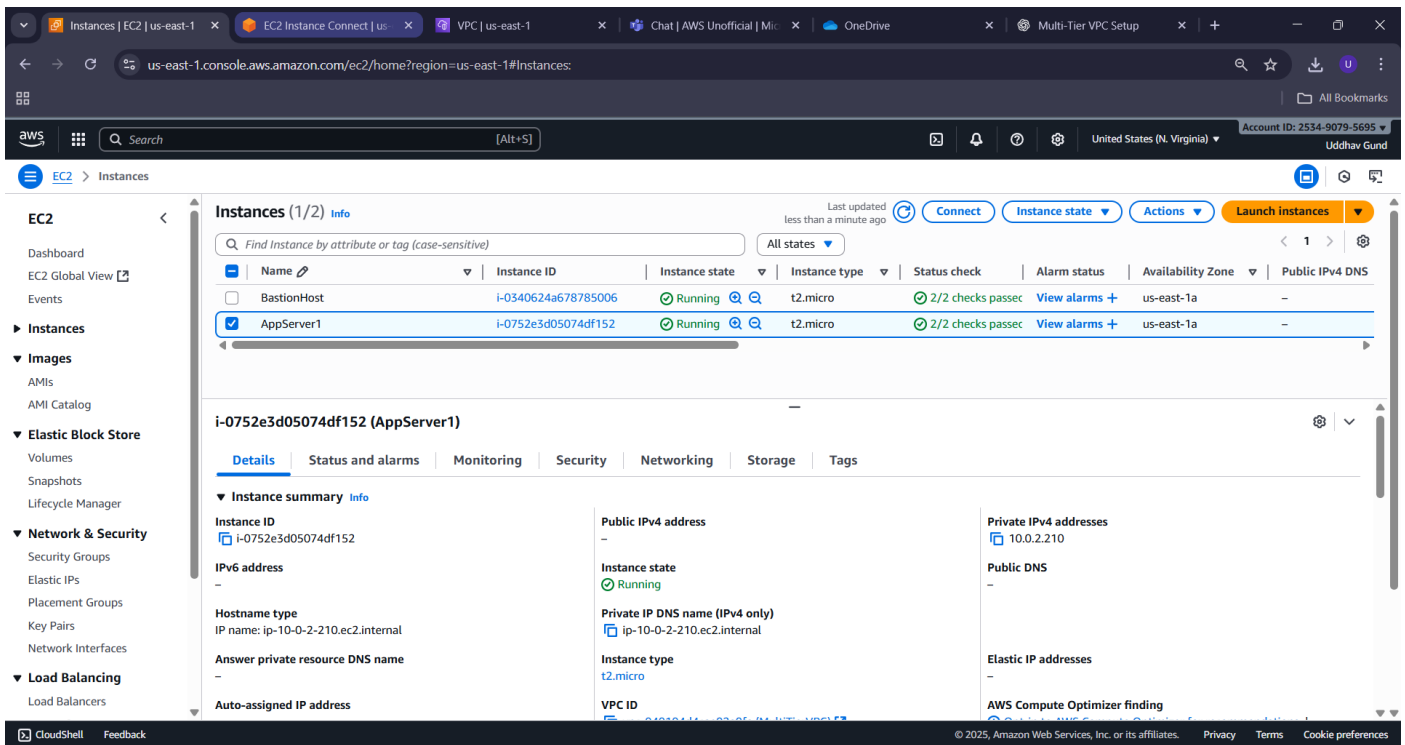
Elastic IP addresses -

AWS Compute Optimizer finding

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Step 7: Launch Private EC2 App Servers

- EC2 → Launch Instance:
 - Name: **AppServer1**
 - Subnet: PrivateSubnet-1a
 - Auto-assign Public IP: Disabled
 - Security Group: AppServerSG
 - Inbound:
 - Allow SSH (22) from BastionSG
 - Allow HTTP (80) from ALBSG
- Outbound: All (default, for NAT access)



Step 8: Test Connectivity with EC2 Instance Connect

1. Go to **EC2** → **Instances**.
 2. Select your **BastionHost**.
 3. Click **Connect** (top-right).
 4. In the pop-up:
 - Choose **EC2 Instance Connect** tab.
 - Username: ec2-user (default for Amazon Linux 2).
 - Leave everything else default.
 5. Click **Connect**.
- ✓ A browser-based terminal will open → you're inside your Bastion host.
- ✓ Connected successfully via **EC2 Instance Connect**.

On terminal ADD BastionHostkey.pem by nano editor:

```
chmod 400 BastionHostkey.pem # set correct permissions if needed
ssh -i BastionHostkey.pem ec2-user@<Bastion Public IP>
[ec2-user@ip-10-0-1-173 ~]$ mkdir ec2
[ec2-user@ip-10-0-1-173 ~]$ nano BastionHostkey.pem
[ec2-user@ip-10-0-1-173 ~]$ chmod 400 BastionHostkey.pem
[ec2-user@ip-10-0-1-173 ~]$ ssh -i BastionHostkey.pem ec2-user@10.0.2.210
```

Your shell prompt now shows:

```
[ec2-user@ip-10-0-2-210 ~]$
```

Test raw IP vs DNS(for check perfect connection)

Inside Private EC2:

```
curl -I http://1.1.1.1
```

```
[ec2-user@ip-10-0-2-210 ~]$ curl -I http://1.1.1.1
HTTP/1.1 301 Moved Permanently
Server: cloudflare
Date: Mon, 08 Sep 2025 07:44:56 GMT
Content-Type: text/html
Content-Length: 167
Connection: keep-alive
Location: https://1.1.1.1/
CF-RAY: 97bcd2ce3c3ed645-IAD
```

Check Public IP your private EC2 uses (via NAT):

```
curl https://checkip.amazonaws.com
```

👉 This should return your NAT's Elastic IP (44.210.184.45).

Test DNS properly:

```
nslookup google.com
```

```
[ec2-user@ip-10-0-2-210 ~]$ curl https://checkip.amazonaws.com
44.210.184.45
[ec2-user@ip-10-0-2-210 ~]$ nslookup google.com
Server:      10.0.0.2
Address:     10.0.0.2#53
```

```
Non-authoritative answer:
Name:   google.com
Address: 192.178.155.138
Name:   google.com
Address: 192.178.155.139
Name:   google.com
Address: 192.178.155.100
Name:   google.com
Address: 192.178.155.101
Name:   google.com
Address: 192.178.155.102
Name:   google.com
Address: 192.178.155.113
Name:   google.com
Address: 2607:f8b0:4004:c1f::66
Name:   google.com
Address: 2607:f8b0:4004:c1f::71
Name:   google.com
Address: 2607:f8b0:4004:c1f::8a
Name:   google.com
Address: 2607:f8b0:4004:c1f::65
```

Deploy simple web app inside Private EC2(AppServer1):

Note:run command one-by-one.

```
sudo yum install -y httpd
```

```
sudo systemctl enable --now httpd
```

```
echo "Hello from Private App Server" | sudo tee /var/www/html/index.html
```

```
curl http://localhost
```

```
us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=us-east-1&connType=standard&instanceId=i-0340624a678785006&osUser=ec2-user&sshPort=22&addressFamily=ipv4

Name: google.com
Address: 2607:f8b0:4004:clif:65

[ec2-user@ip-10-0-2-210 ~]$ sudo yum install -y httpd
Amazon Linux 2023 repository
Amazon Linux 2023 Kernel Livepatch repository
Dependencies resolved.

Package Architecture Version Repository Size
-----
Installing:
httpd x86_64 2.4.64-1.amzn2023.0.1 amazonlinux 47 k
Installing dependencies:
apr x86_64 1.7.5-1.amzn2023.0.4 amazonlinux 129 k
apr-util x86_64 1.6.3-1.amzn2023.0.1 amazonlinux 98 k
generic-logos-httpd noarch 18.0.0-12.amzn2023.0.3 amazonlinux 19 k
httpd-core x86_64 2.4.64-1.amzn2023.0.1 amazonlinux 1.4 M
httpd-filesystem noarch 2.4.64-1.amzn2023.0.1 amazonlinux 13 k
httpd-tools x86_64 2.4.64-1.amzn2023.0.1 amazonlinux 81 k
libbrotli x86_64 1.0.9-4.amzn2023.0.2 amazonlinux 315 k
mailcap noarch 2.1.49-3.amzn2023.0.3 amazonlinux 33 k
Installing weak dependencies:
apr-util-openssl x86_64 1.6.3-1.amzn2023.0.1 amazonlinux 17 k
mod_http2 x86_64 2.0.27-1.amzn2023.0.3 amazonlinux 166 k
mod_lua x86_64 2.4.64-1.amzn2023.0.1 amazonlinux 60 k

Transaction Summary
-----
Install 12 Packages
Total download size: 2.3 M

i-0340624a678785006 (BastionHost)
PublicIPs: 100.24.28.80 PrivateIPs: 10.0.1.173

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```

```
us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=us-east-1&connType=standard&instanceId=i-0340624a678785006&osUser=ec2-user&sshPort=22&addressFamily=ipv4

Installing : httpd-2.4.64-1.amzn2023.0.1.x86_64 12/12
Running scriptlet: httpd-2.4.64-1.amzn2023.0.1.x86_64 12/12
Verifying : apr-1.7.5-1.amzn2023.0.4.x86_64 1/12
Verifying : apr-util-1.6.3-1.amzn2023.0.1.x86_64 2/12
Verifying : apr-util-openssl-1.6.3-1.amzn2023.0.1.x86_64 3/12
Verifying : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch 4/12
Verifying : httpd-2.4.64-1.amzn2023.0.1.x86_64 5/12
Verifying : httpd-core-2.4.64-1.amzn2023.0.1.x86_64 6/12
Verifying : httpd-filesystem-2.4.64-1.amzn2023.0.1.noarch 7/12
Verifying : httpd-tools-2.4.64-1.amzn2023.0.1.x86_64 8/12
Verifying : libbrotli-1.0.9-4.amzn2023.0.2.x86_64 9/12
Verifying : mailcap-2.1.49-3.amzn2023.0.3.noarch 10/12
Verifying : mod_http2-2.0.27-1.amzn2023.0.3.x86_64 11/12
Verifying : mod_lua-2.4.64-1.amzn2023.0.1.x86_64 12/12

Installed:
apr-1.7.5-1.amzn2023.0.4.x86_64 apr-util-1.6.3-1.amzn2023.0.1.x86_64
apr-util-openssl-1.6.3-1.amzn2023.0.1.x86_64 generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch
httpd-2.4.64-1.amzn2023.0.1.x86_64 httpd-core-2.4.64-1.amzn2023.0.1.x86_64
httpd-filesystem-2.4.64-1.amzn2023.0.1.noarch httpd-tools-2.4.64-1.amzn2023.0.1.x86_64
libbrotli-1.0.9-4.amzn2023.0.2.x86_64 mailcap-2.1.49-3.amzn2023.0.3.noarch
mod_http2-2.0.27-1.amzn2023.0.3.x86_64 mod_lua-2.4.64-1.amzn2023.0.1.x86_64

Complete!
[ec2-user@ip-10-0-2-210 ~]$ sudo systemctl enable --now httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
[ec2-user@ip-10-0-2-210 ~]$ echo "Hello from Private App Server" | sudo tee /var/www/html/index.html
Hello from Private App Server
[ec2-user@ip-10-0-2-210 ~]$ curl http://localhost
Hello from Private App Server
[ec2-user@ip-10-0-2-210 ~]$
```

Step 9: Add ALB in Public Subnet

1. Security Group for ALB

- Create ALB-SG
 - Inbound:
 - HTTP (80) from 0.0.0.0/0 (or HTTPS if you want SSL)
 - Outbound: All traffic (default).

2. Security Group for App Servers

- Update AppServer-SG
 - Inbound:
 - SSH (22) from **Bastion-SG** (already done)
 - HTTP (80) from **ALB-SG** (new)
 - Outbound: All traffic (default).

3. Create Target Group

1. Go to **EC2** → **Target Groups** → **Create Target Group**.
2. Type: **Instances**
3. Name: **AppServersTG**
4. Protocol: **HTTP**, Port **80**
5. VPC: **MultiTierVPC**
6. Health check: **/** (path to test)
7. Register your **Private EC2 App Servers (10.0.2.x)** as targets.

The screenshot displays the AWS Management Console for the 'AppServersTG' Target Group. The left sidebar shows the navigation menu with categories like Images, Elastic Block Store, Network & Security, Load Balancing, and Auto Scaling. The main content area is titled 'AppServersTG' and includes an 'Actions' button. Below the title, there's a 'Details' section with a table of attributes: Target type (Instance), IP address type (IPv4), Protocol : Port (HTTP: 80), Load balancer (AppALB), Protocol version (HTTP1), and VPC (vpc-040104d4cae02a0fa). A summary row shows 1 Total target, 1 Healthy, 0 Unhealthy, 0 Unused, 0 Initial, and 0 Draining. Below this is a 'Distribution of targets by Availability Zone (AZ)' section. The 'Targets' tab is active, showing a table of 'Registered targets (1/1)'. The table has columns for Instance ID, Name, Port, Zone, Health status, Health status details, Admini..., Overri..., Launch..., and Anomaly detectio... The single target listed is 'i-0752e3d05074df152' with name 'AppServer1', port '80', zone 'us-east-1a (us...)', and health status 'Healthy'.

4. Create Application Load Balancer

1. Go to **EC2** → **Load Balancers** → **Create Load Balancer**.
2. Choose **Application Load Balancer**.
3. Name: **AppALB**
4. Scheme: **Internet-facing**
5. VPC: **MultiTierVPC**
6. Subnets: Select both **Public Subnets** (if you only have 1 now, select it).
7. Security Group: Select **ALB-SG**.
8. Listener:
 - o Protocol: **HTTP (80)**
 - o Forward to: **AppServersTG**

Create application load balancerEC2 Instance Connect | us-east-1Instances | EC2 | us-east-1(1) Chat | Smruti Kare | MicrosoftOneDriveus-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#CreateALBWizard:All Bookmarks

naT gatewaysUnited States (N. Virginia)Account ID: 2534-9079-5695Uddhav Gund

EC2Load balancersCreate Application Load Balancer

Create Application Load Balancer

The Application Load Balancer distributes incoming HTTP and HTTPS traffic across multiple targets such as Amazon EC2 instances, microservices, and containers, based on request attributes. When the load balancer receives a connection request, it evaluates the listener rules in priority order to determine which rule to apply, and if applicable, it selects a target from the target group for the rule action.

► How Application Load Balancers work

Basic configuration

Load balancer name
Name must be unique within your AWS account and can't be changed after the load balancer is created.

AppALB

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Scheme | Info
Scheme can't be changed after the load balancer is created.

☒ **Internet-facing**

- Serves internet-facing traffic.
- Has public IP addresses.
- DNS name resolves to public IPs.
- Requires a public subnet.

☐ **Internal**

- Serves internal traffic.
- Has private IP addresses.
- DNS name resolves to private IPs.
- Compatible with the IPv4 and Dualstack IP address types.

Load balancer IP address type | Info
Select the front-end IP address type to assign to the load balancer. The VPC and subnets mapped to this load balancer must include the selected IP address types. Public IPv4 addresses have an additional cost.

☒ **IPv4**

Includes only IPv4 addresses.

☐ **Dualstack**

Includes IPv4 and IPv6 addresses.

☐ **Dualstack without public IPv4**

Includes a public IPv6 address, and private IPv4 and IPv6 addresses. Compatible with **Internet-facing** load balancers only.

Network mapping

The load balancer routes traffic to targets in the selected subnets, and in accordance with your IP address settings.

VPC | Info
The load balancer will exist and scale within the selected VPC. The selected VPC is also where the load balancer targets must be hosted unless routing to Lambda or on-premises targets, or if using VPC peering. To confirm the VPC for your targets, view [target groups](#).

vpc-04010d4cae02a0fa (MultiTierVPC)
10.0.0.0/16

[Create VPC](#)

IP pools | Info
You can optionally choose to configure an IPAM pool as the preferred source for your load balancers IP addresses. Create or view Pools in the [Amazon VPC IP Address Manager console](#).

☐ **Use IPAM pool for public IPv4 addresses**
The IPAM pool you choose will be the preferred source of public IPv4 addresses. If the pool is depleted IPv4 addresses will be assigned by AWS.

Availability Zones and subnets | Info
Select at least two Availability Zones and a subnet for each zone. A load balancer node will be placed in each selected zone and will automatically scale in response to traffic. The load balancer routes traffic to targets in the selected Availability Zones only.

☒ **us-east-1a (use1-az6)**

Subnet
Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

subnet-051194440aa334ca3
IPv4 subnet CIDR: 10.0.1.0/24

PublicSubnet-1a

☒ **us-east-1b (use1-az1)**

Subnet
Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

subnet-0c8a10789db0fb423
IPv4 subnet CIDR: 10.0.3.0/24

PublicSubnet-1b

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Create application load balancerEC2 Instance Connect | us-east-1VPC | us-east-1(1) Chat | Smruti Kare | MicrosoftOneDriveus-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#CreateALBWizard:All Bookmarks

Search[Alt+S]United States (N. Virginia)Account ID: 2534-9079-5695Uddhav Gund

EC2Load balancersCreate Application Load Balancer

☐ Dualstack without public IPv4
Includes a public IPv6 address, and private IPv4 and IPv6 addresses. Compatible with **Internet-facing** load balancers only.

Network mapping

The load balancer routes traffic to targets in the selected subnets, and in accordance with your IP address settings.

VPC | Info
The load balancer will exist and scale within the selected VPC. The selected VPC is also where the load balancer targets must be hosted unless routing to Lambda or on-premises targets, or if using VPC peering. To confirm the VPC for your targets, view [target groups](#).

vpc-04010d4cae02a0fa (MultiTierVPC)
10.0.0.0/16

[Create VPC](#)

IP pools | Info
You can optionally choose to configure an IPAM pool as the preferred source for your load balancers IP addresses. Create or view Pools in the [Amazon VPC IP Address Manager console](#).

☐ **Use IPAM pool for public IPv4 addresses**
The IPAM pool you choose will be the preferred source of public IPv4 addresses. If the pool is depleted IPv4 addresses will be assigned by AWS.

Availability Zones and subnets | Info
Select at least two Availability Zones and a subnet for each zone. A load balancer node will be placed in each selected zone and will automatically scale in response to traffic. The load balancer routes traffic to targets in the selected Availability Zones only.

☒ **us-east-1a (use1-az6)**

Subnet
Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

subnet-051194440aa334ca3
IPv4 subnet CIDR: 10.0.1.0/24

PublicSubnet-1a

☒ **us-east-1b (use1-az1)**

Subnet
Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

subnet-0c8a10789db0fb423
IPv4 subnet CIDR: 10.0.3.0/24

PublicSubnet-1b

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us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#CreateALBWizard:

Security groups Info

A security group is a set of firewall rules that control the traffic to your load balancer. Select an existing security group, or you can [create a new security group](#).

Security groups

Select up to 5 security groups

ALB-SG

Listeners and routing Info

A listener is a process that checks for connection requests using the port and protocol you configure. The rules that you define for a listener determine how the load balancer routes requests to its registered targets.

Listener HTTP:80

Protocol HTTP Port 80

Default action Forward to AppServersTG Target type: Instance, IPv4

Listener tags - optional

Consider adding tags to your listener. Tags enable you to categorize your AWS resources so you can more easily manage them.

Add listener tag

Add listener

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#LoadBalancerloadBalancerArn=arn:aws:elasticloadbalancing:us-east-1:253490795695:loadbalancer/app/AppALB/027c03e9f99a44...

AppALB

Details

Load balancer type Application Status Active VPC vpc-040104d4cae02a0fa Load balancer IP address type IPv4

Scheme Internet-facing Hosted zone Z35SXDOTRQ7X7K Availability Zones subnet-0c8a10789db0fb423 us-east-1b (use 1-az1) subnet-051194440aa334ca3 us-east-1a (use 1-az6) Date created September 8, 2025, 13:45 (UTC+05:30)

Load balancer ARN arn:aws:elasticloadbalancing:us-east-1:253490795695:loadbalancer/app/AppALB/027c03e9f99a446e DNS name info AppALB-1302416658.us-east-1.elb.amazonaws.com (A Record)

Listeners and rules

Listeners and rules (1) Info

A listener checks for connection requests on its configured protocol and port. Traffic received by the listener is routed according to the default action and any additional rules.

Protocol:Port	Default action	Rules	ARN	Security policy	Default SSL/TLS certificate	mTLS	Trust store
HTTP:80	Forward to target group AppServersTG 1 (100%) Target group stickiness: Off	1 rule	ARN	Not applicable	Not applicable	Not applicable	Not applicable

5. Test the Setup

1. Inside Private EC2, confirm Apache is running:
2. `curl http://localhost`

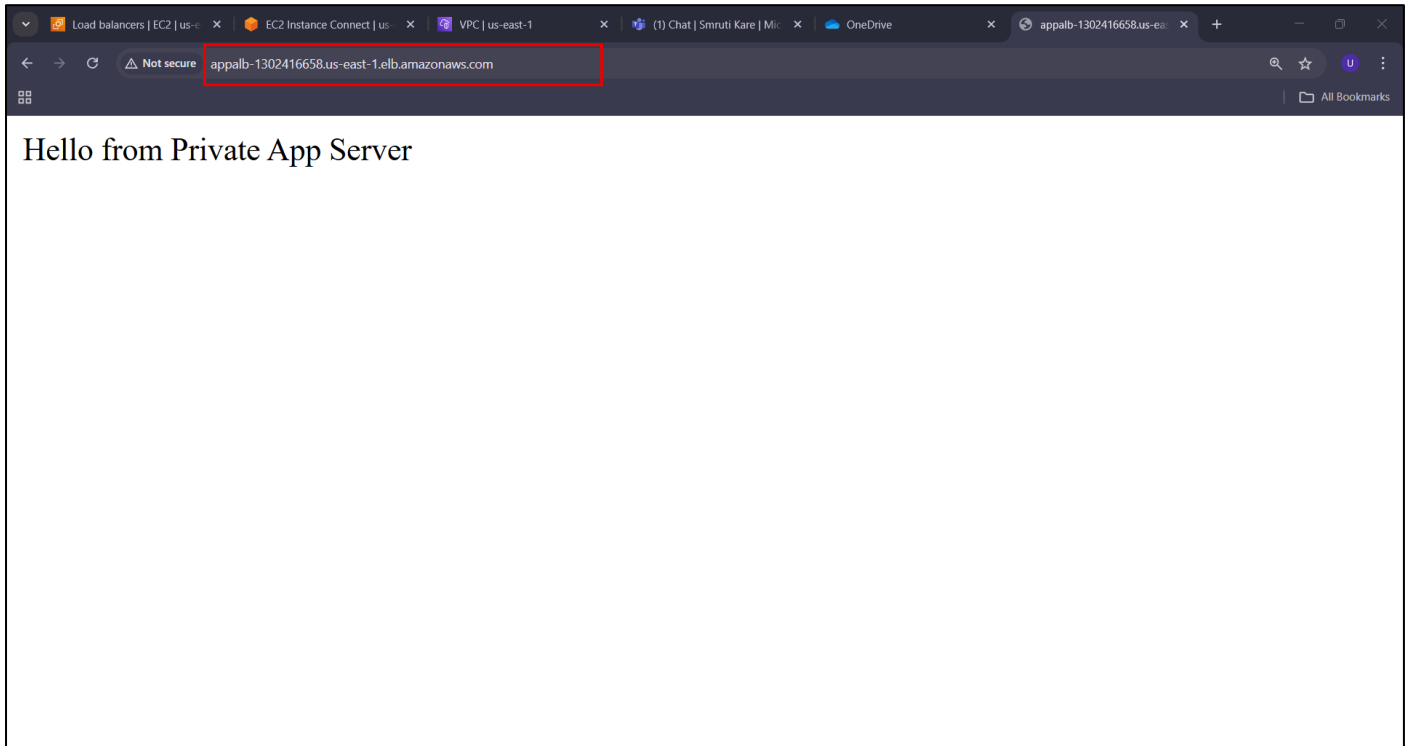
Should show `Hello from Private App Server.`

```
[ec2-user@ip-10-0-2-210 ~]$ curl http://localhost
Hello from Private App Server
[ec2-user@ip-10-0-2-210 ~]$
```

i-0340624a678785006 (BastionHost)

PublicIPs: 100.24.28.80 PrivateIPs: 10.0.1.173

3. open or copy and paste on browser the **ALB DNS name** (shown in ALB details).



👉 Result : `Hello from Private App Server` 🎉

✅ Final Architecture

- **Public Subnet:** Bastion Host + ALB + NAT Gateway
- **Private Subnet:** App Servers (no public IPs)
- **Users:** Access app only through ALB
- **Admins:** SSH to App Servers only via Bastion